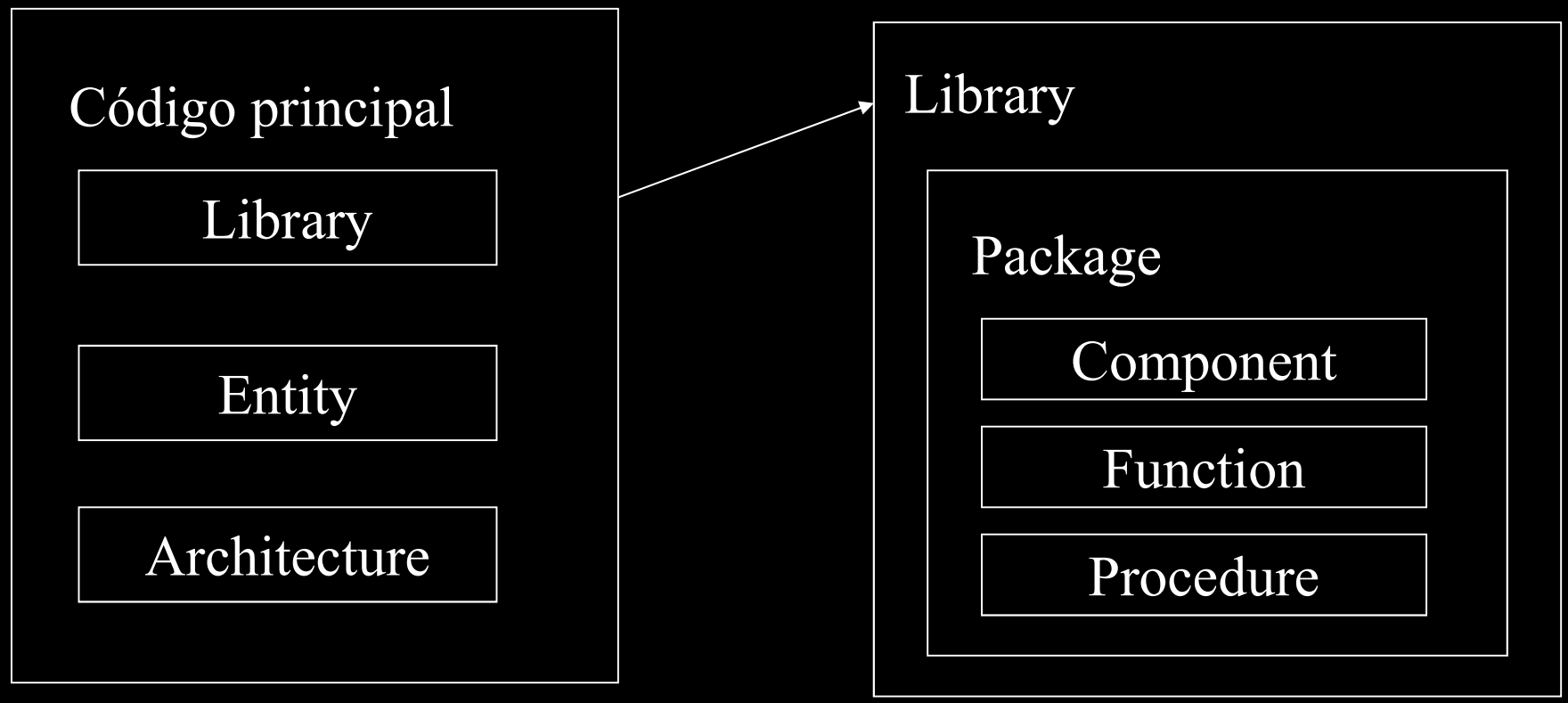
Circuitos Digitais I

VHDL: Packages Function, Procedure e Component

Circuitos Digitais I

VHDL: Packages and Components



Circuitos Digitais I

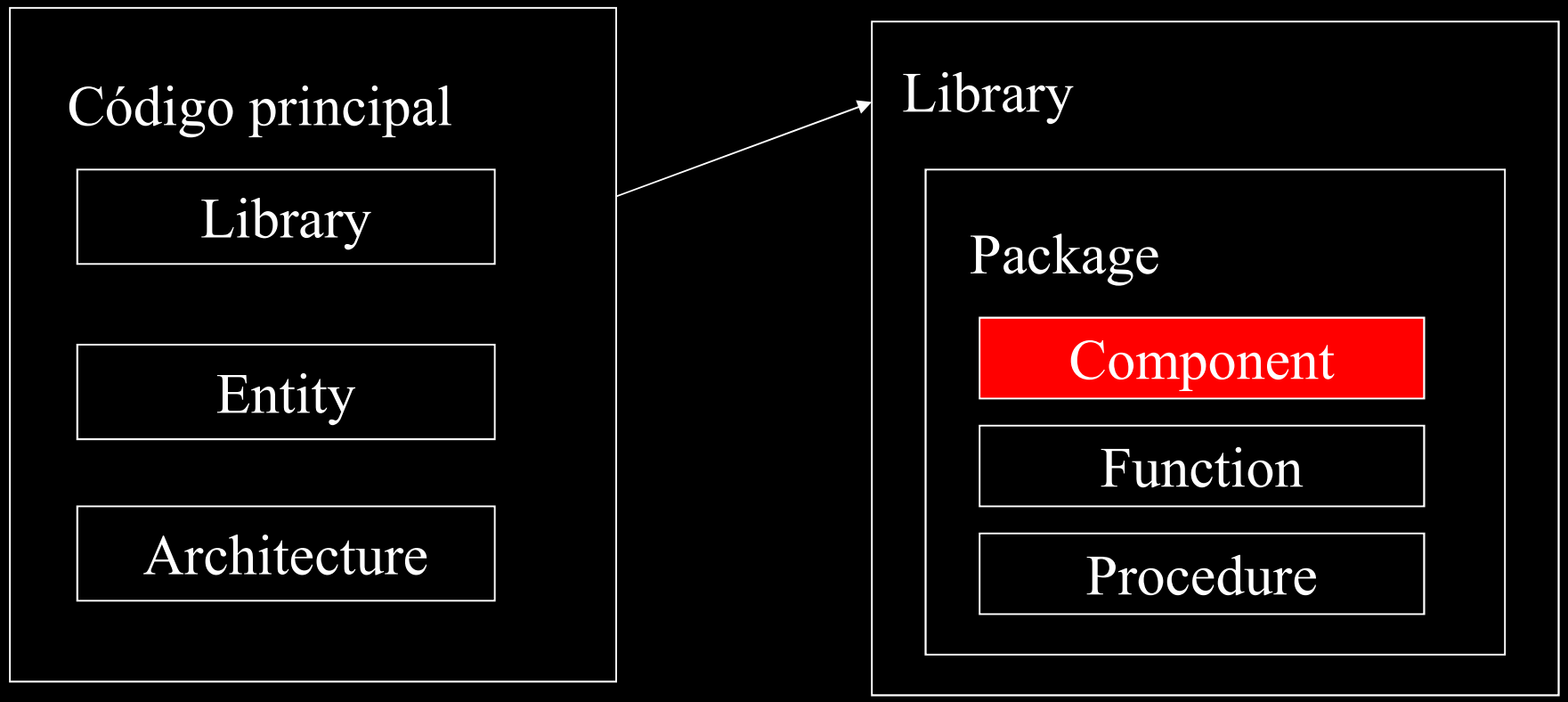
VHDL: Package

- Fragmentos de código escritos na forma de COMPONENTS, FUNCTIONS, or PROCEDURES são inseridos em um PACKAGE e compilados em uma LIBRARY.
- Recurso para propiciar particionamento e reuso

```
PACKAGE nome_pacote IS
    .
    .(declarações)
END nome_pacote;
[PACKAGE BODY nome_pacote IS
   (implementacão funções e procedimentos)
END nome_pacote;]
```

Circuitos Digitais I

VHDL: Package com Components



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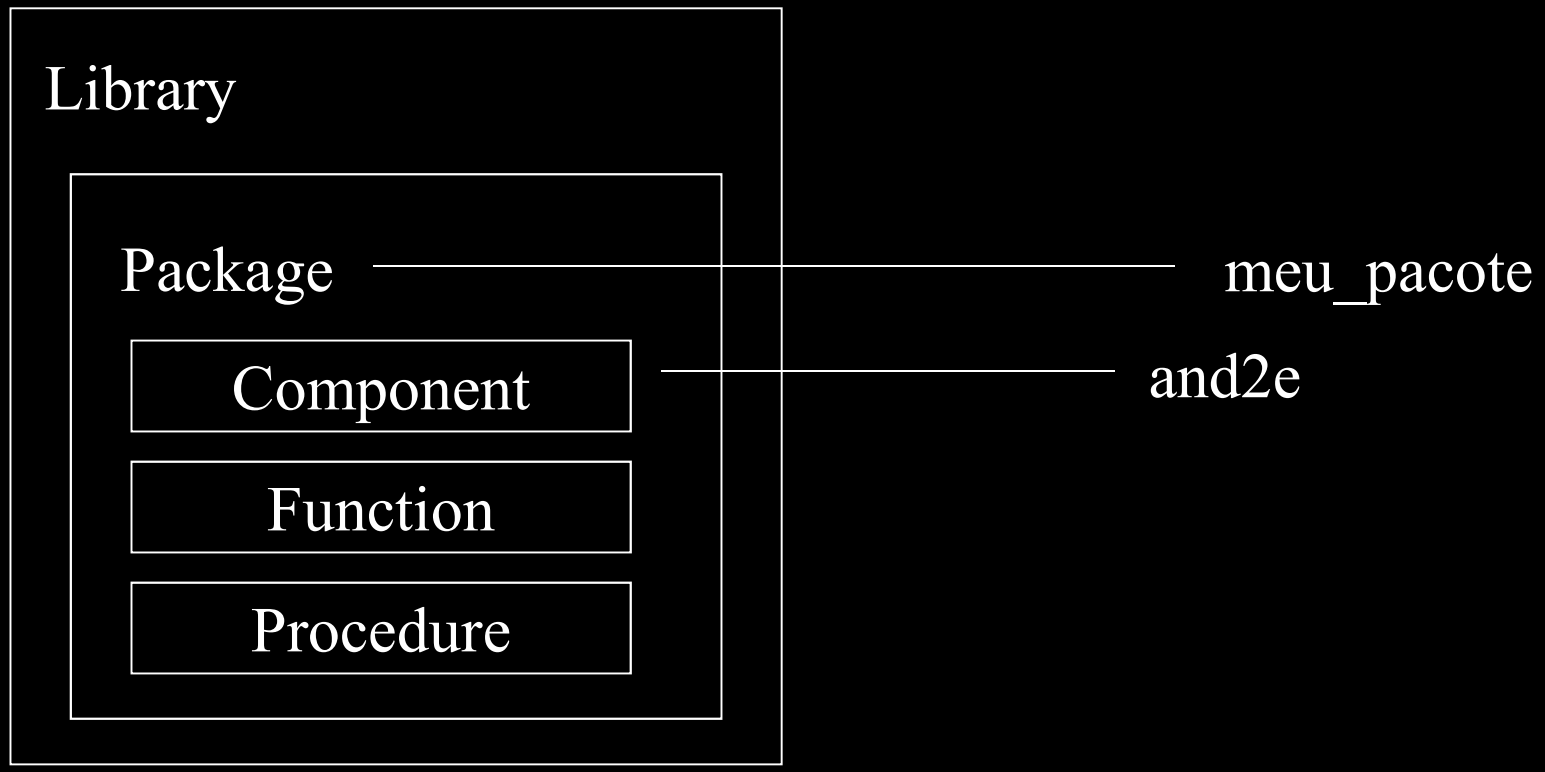
VHDL: Package com Components



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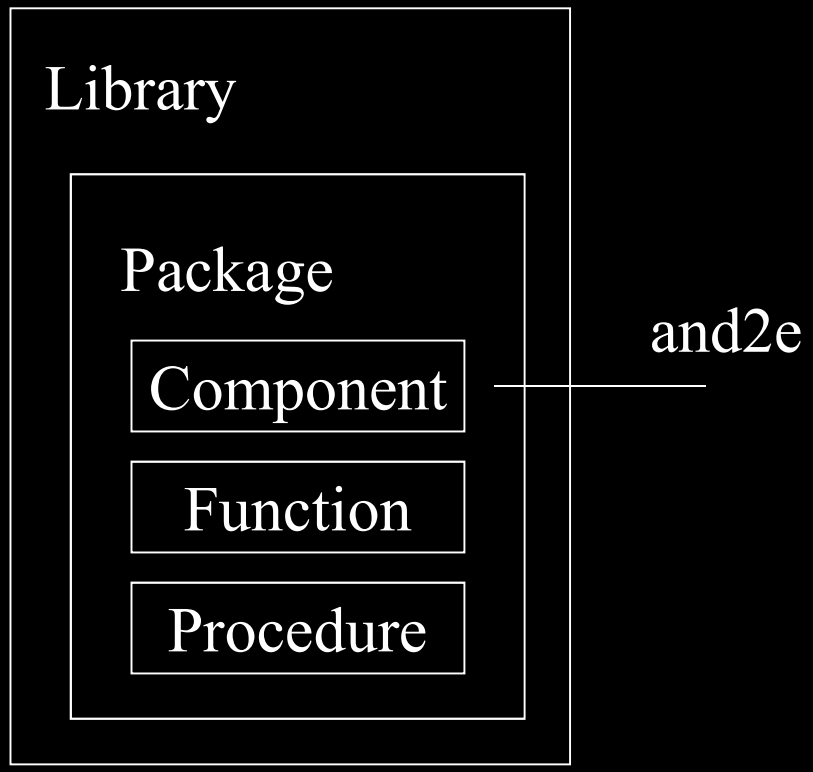
VHDL: Package com Components



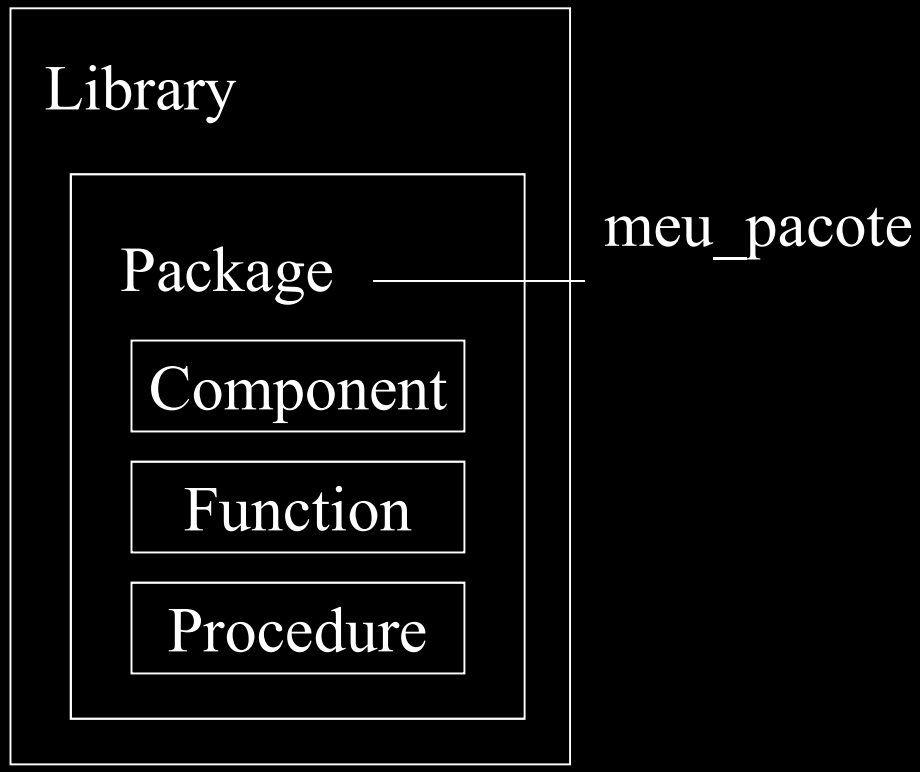

```
entity and2e is
        port
        (
         e1, e2: IN STD_LOGIC;
         s1: OUT STD_LOGIC
        );
end and2e;

ARCHITECTURE fluxodados
of and2e is
 begin
   s1 <= e1 and e2;
end fluxodados;
```

arquivo:

and2e.vhd

Circuitos Digitais I

VHDL: Package com Components




```
PACKAGE meu_pacote IS

  component and2e
    port(
        e1, e2: IN  STD_LOGIC;
        s1: OUT STD_LOGIC
         );
  end component;
end meu_pacote;
```

arquivo:

meu_pacote.vhd

Circuitos Digitais I

VHDL: Package com Components

arquivo:
pacote_componentes.vhd

```
USE work.meu_pacote.all;

entity pacotes_componentes is
        port
        (
         en1,en2: IN STD_LOGIC;
         sa1: OUT STD_LOGIC
        );
end pacotes_componentes;

ARCHITECTURE fluxodados of
pacotes_componentes is
 begin
   bloco1: and2e PORT MAP (en1, en2, sa1);
end fluxodados;
```

Circuitos Digitais I

VHDL: Package com Components

pacote_componentes.vhd
USE work.**meu_pacote**.all;
.
.
.
.

Library **meu_pacote**

Package **meu_pacote**

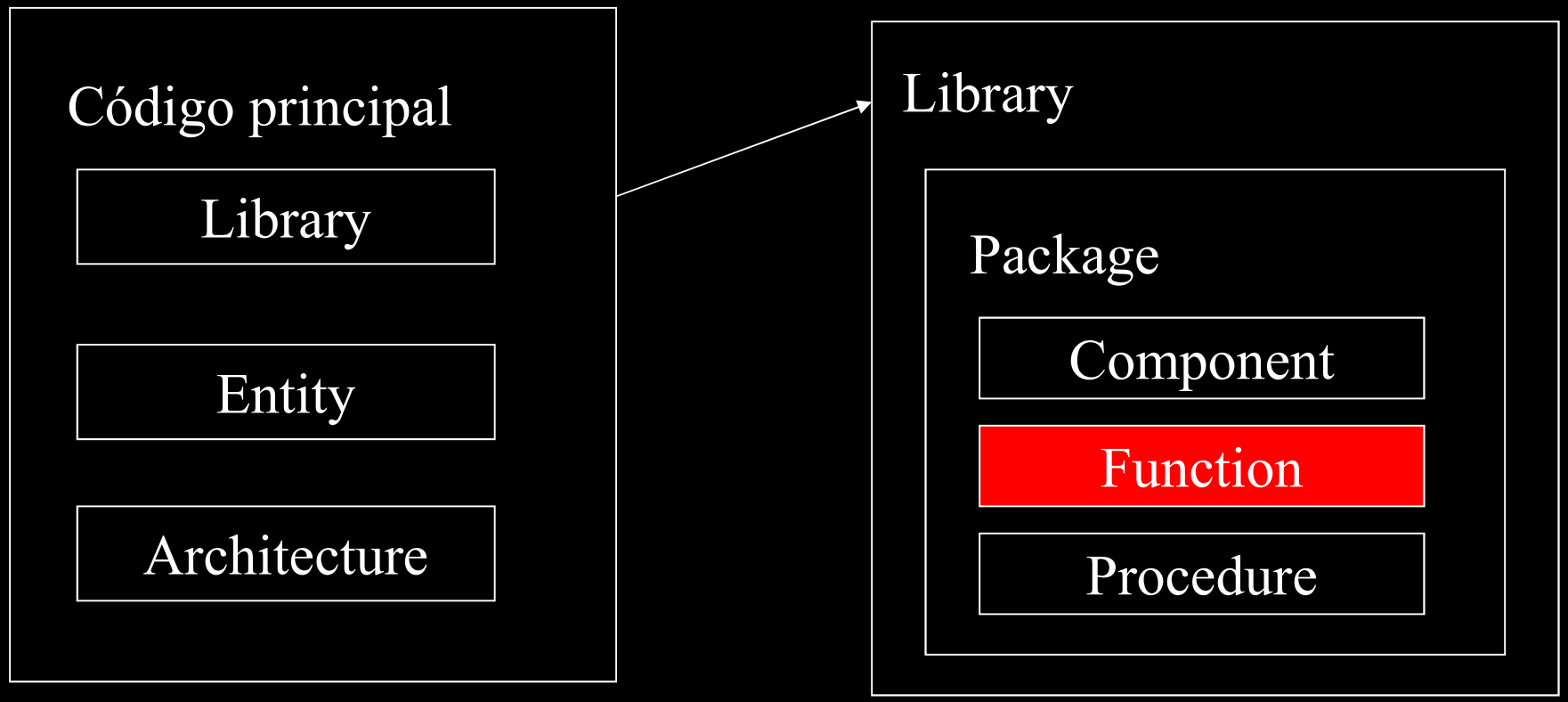
Component **and2e**

Function

Procedure

Circuitos Digitais I

VHDL: Package com Function



Circuitos Digitais I

VHDL: Package com Function

•Function

• FUNCTION é uma seção de código sequencial

• Similar a operação de PROCESS

• palavras IF, CASE, LOOP são permitidas em FUNCTION

• Uma função compreende a declaração da mesma com a implementação e a chamada a essa função

Circuitos Digitais I

VHDL: Package com Function

Function: sintaxe

```
FUNCTION nome_funcao [<parameter list>] RETURN
tipo_dados IS
[declaracoes]
BEGIN
.
.
código
END nome_funcao;
```

Circuitos Digitais I

VHDL: Package com Function

Function: operação AND com três entradas

```
FUNCTION and_3_entradas(SIGNAL a,b,c: STD_LOGIC)
RETURN STD_LOGIC IS
    begin
      RETURN (a and b and c);
    end and_3_entradas;
```

Circuitos Digitais I

VHDL: Package com Function

```
PACKAGE meu_pacote IS

end meu_pacote;

PACKAGE BODY meu_pacote IS

FUNCTION and_3_entradas(SIGNAL a,b,c: STD_LOGIC)
RETURN STD_LOGIC IS
    begin
      RETURN (a and b and c);
    end and_3_entradas;

end meu_pacote;
```

Circuitos Digitais I

VHDL: Package com Function

```
USE work.meu_pacote.all;

entity pacotes_componentes is
        port
        (
         en1,en2,en3: IN STD_LOGIC;
         sa1: OUT STD_LOGIC
        );
end pacotes_componentes;

ARCHITECTURE fluxodados of pacotes_componentes is
 begin
   sa1 <= and_3_entradas(en1,en2,en3);
end fluxodados;
```

Circuitos Digitais I

VHDL: Package com Function

```
pacote_componentes.vhd
USE work.meu_pacote.all;
.
x <=  and_3_entradas(en1,en2,en3);
.
```

Library **meu_pacote**

Package **meu_pacote**

- Component
- Function **and_3_entradas**
- Procedure

Circuitos Digitais I

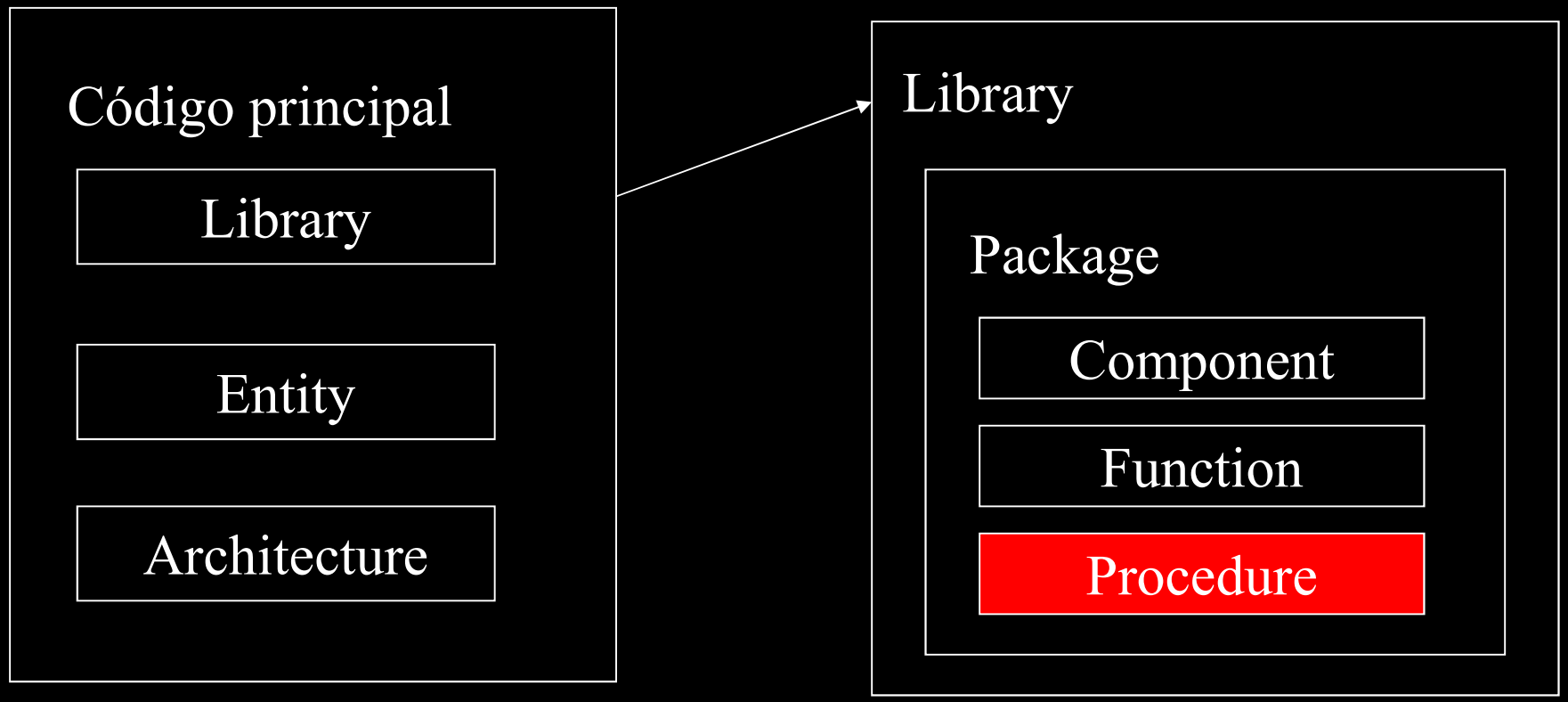
VHDL: Function localização

Para efeito de reutilização FUNCTION e PROCEDURE geralmente estão inseridos em PACKAGES, contudo podem estar inseridos no código principal declarados em ARCHITECTURE ou ENTITY

```
ENTITY exemplo IS
.
 END exemplo;
ARCHITECTURE arquit OF exemplo IS
FUNCTION outro_and(SIGNAL a,b: STD_LOGIC)  RETURN BOOLEAN IS
  BEGIN
     RETURN a and b;
  END outro_and;
BEGIN
.
END arquit;
```

Circuitos Digitais I

VHDL: Package com Procedure



VHDL: PROCEDURE

A PROCEDURE e FUNCTION tem os mesmos propósitos, contudo PROCEDURE pode retornar mais de um valor.

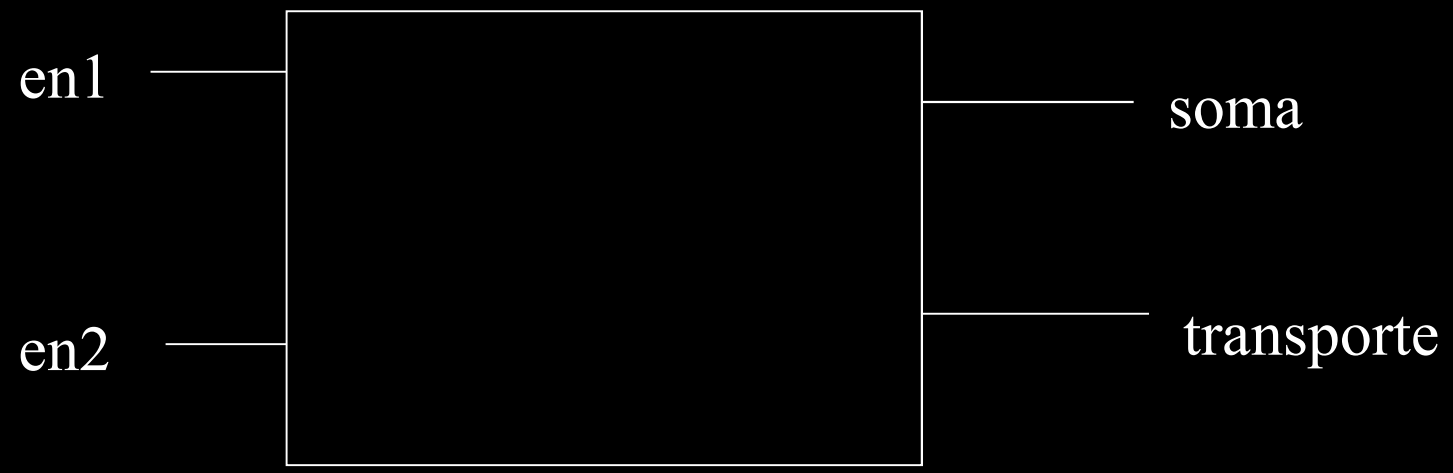
Sintaxe:

```
PROCEDURE procedure_name [<parameter list>] IS
[declarations]
BEGIN
(sequential statements)
END procedure_name;
```

Circuitos Digitais I

VHDL: PACKAGE PROCEDURE

Procedure: somador dois bits com transporte



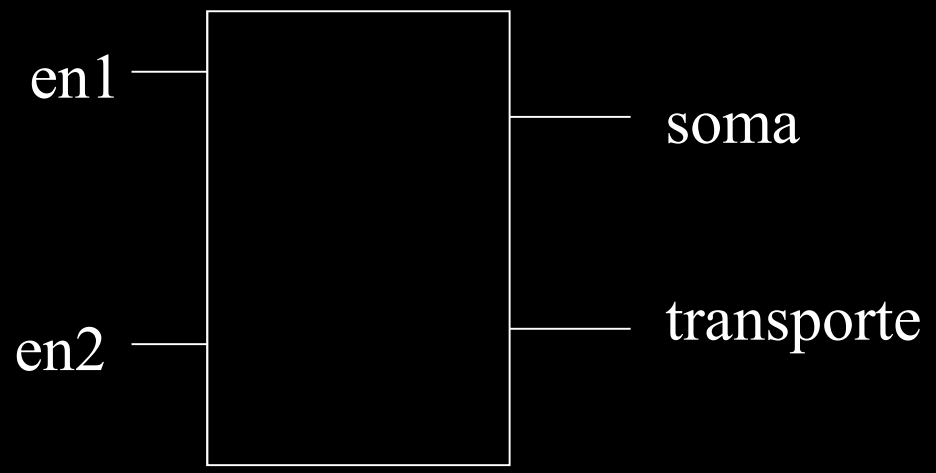
Circuitos Digitais I

VHDL: PACKAGE PROCEDURE

Procedure: somador dois bits com transporte

Tabela verdade

en1	en2	soma	transporte
0	0	0	0
1	0	1	0
1	1	0	1



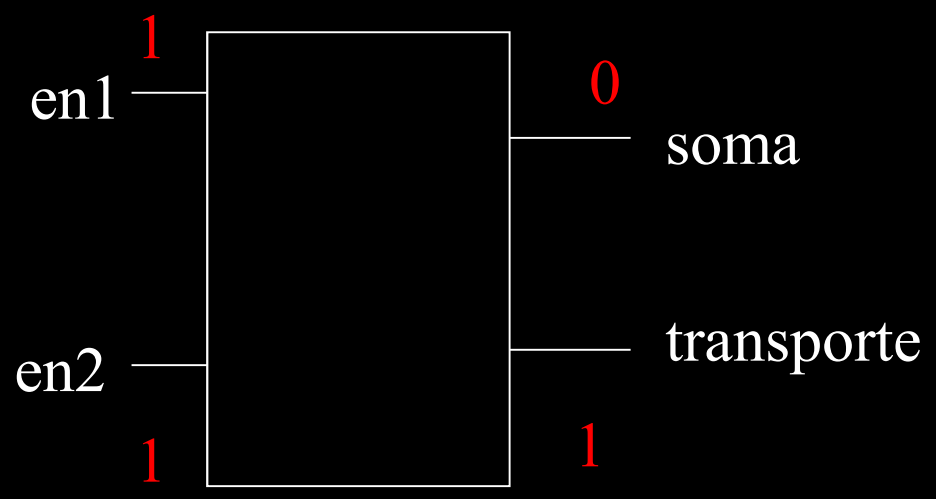
Circuitos Digitais I

VHDL: PACKAGE PROCEDURE

Procedure: somador dois bits com transporte

Tabela verdade:

en1	en2	soma	transporte
0	0	0	0
1	0	1	0
1	1	0	1



Operação 1 + 1

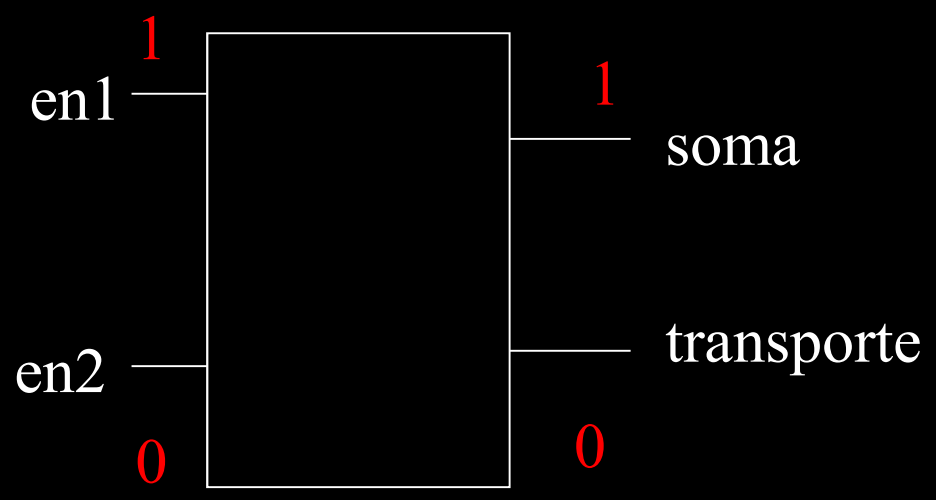
Circuitos Digitais I

VHDL: PACKAGE PROCEDURE

Procedure: somador dois bits com transporte

Tabela verdade:

en1	en2	soma	transporte
0	0	0	0
1	0	1	0
1	1	0	1



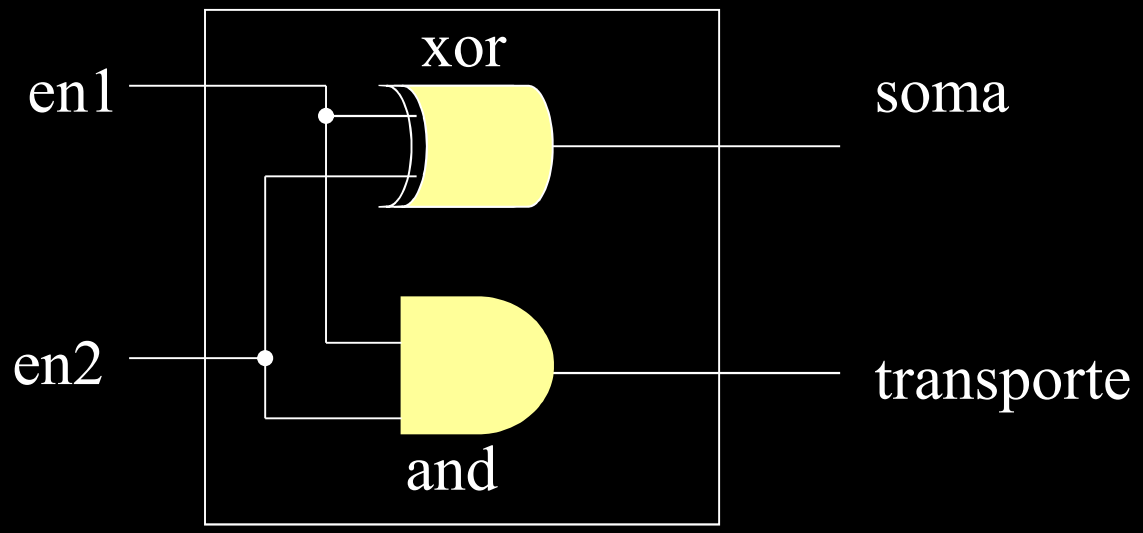
Operação 1 + 0

Circuitos Digitais I

VHDL: PACKAGE PROCEDURE

Procedure: somador dois bits com transporte

Circuito com Blocos Lógicos

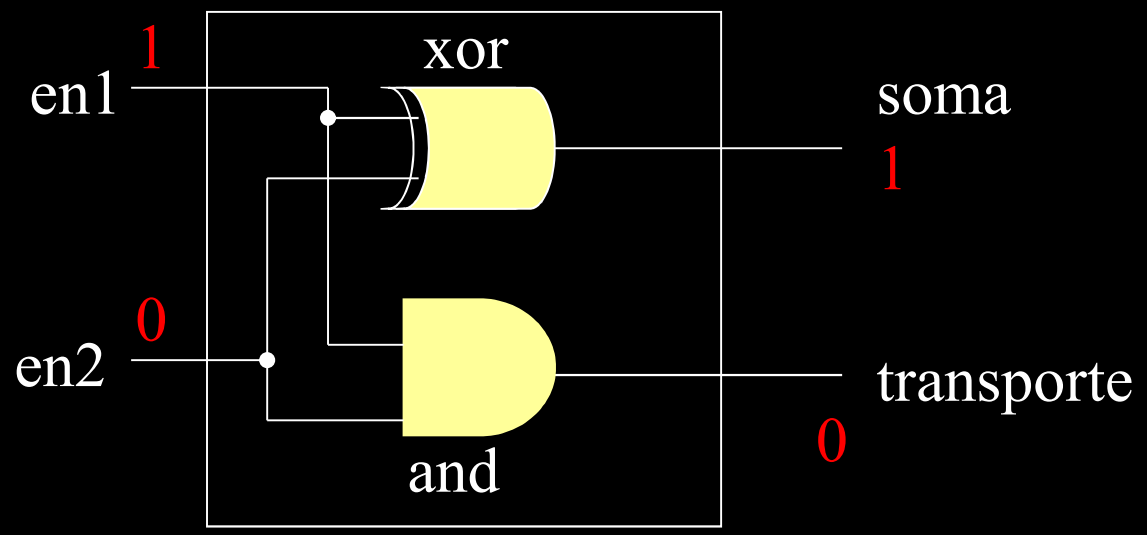


Circuitos Digitais I

VHDL: PACKAGE PROCEDURE

Procedure: somador dois bits com transporte

Circuito com Blocos Lógicos



Xor en1 =1, en2 =0 , soma, = 1

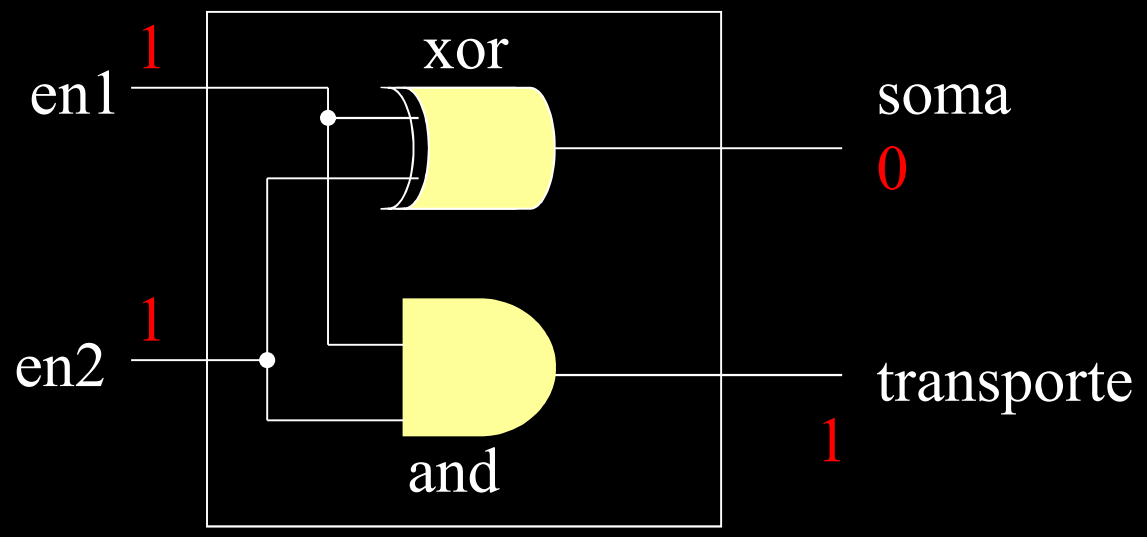
And en1 =1, en2=0, transporte = 0

Circuitos Digitais I

VHDL: PACKAGE PROCEDURE

Procedure: somador dois bits com transporte

Circuito com Blocos Lógicos



Xor en1 =1, en2 =1 , soma, = 0

And en1 =1, en2=1, transporte = 1

Circuitos Digitais I

VHDL: PACKAGE PROCEDURE

Procedure: somador dois bits com transporte

```
PROCEDURE soma_2_bits(SIGNAL a,b:IN  STD_LOGIC;
SIGNAL soma,transporte: OUT STD_LOGIC ) is

 BEGIN
     soma <= a xor b;
     transporte <= a and b;
 END soma_2_bits;
```

Circuitos Digitais I

VHDL: Package com Procedure

```
pacote_componentes.vhd
USE work.meu_pacote.all;
.
port
   (
   en1,en2: IN STD_LOGIC;
   soma,transporte: OUT STD_LOGIC
   );
.
Soma_2_bits(en1,en2  soma, transporte);
.
```

Library **meu_pacote**

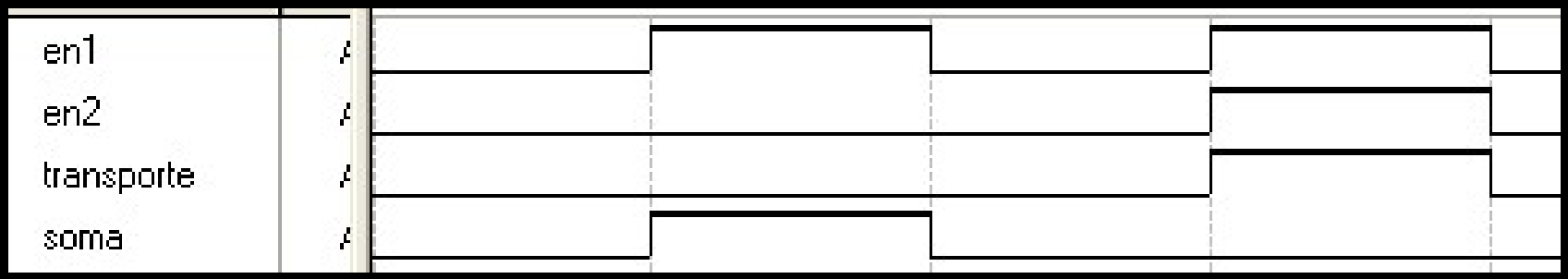
Package **meu_pacote**

- Component
- Function
- Procedure **soma_2_bits**

Circuitos Digitais I

VHDL: Package com Procedure

Somador_2_bits – Simulação Quartus



En1 = 0	En1 = 1	En1 = 1
En2 = 0	En2 = 0	En2 = 1
Transporte =0	Transporte =0	Transporte =1
Soma = 0	Soma = 1	Soma = 0

Circuitos Digitais I

VHDL: Procedure localização

PROCEDURE geralmente esta inseridos em PACKAGES, contudo pode estar inseridos no código principal declarados em ARCHITECTURE ou ENTITY

```
ENTITY exemplo IS
.
 END exemplo;
ARCHITECTURE arquit OF exemplo IS
Procedure outro_or(SIGNAL a,b:IN STD_LOGIC; SIGNAL  s: OUT
STD_LOGIC) is
   BEGIN
       s <= a or b;
   END outro_or;
BEGIN
.
END arquit;
```